

10	Toners	Compatible with printers in use	2 pcs	13:1
11	Internet	Broadband or Wi-Fi for research and online tools	1 connection	Shared
D	Tools and Equipment			
12	Projectors	For displaying teaching materials and multimedia content	1 unit	25:1
13	Printers	For printing training materials and trainee documents	4 units	6:1
14	Computers / Smartphones	For simulations, research, online assessments	25 units	1:1

INFORMATION SYSTEMS DEVELOPMENT

ISCED UNIT CODE: 0613 554 11A

UNIT CODE: ICT/CU/CS/CR/06/6/MA

Relationship to Occupational Standards

This unit addresses the unit of competency: Develop Information System

Duration of Unit: 200 hours

Unit Description

This unit covers the competencies required to develop an information system. It involves understanding fundamentals of information systems, applying security measures to data, hardware, software in automated environment, understanding the software development process, demonstrating human computer interaction principles, understanding the VB.net programming environment and developing and testing a VB.NET application.

Summary of Learning Outcomes

Learning outcomes	Duration (hours)
1. Fundamentals of Information Systems	30
2. Security measures to data, hardware, software in automated environment	20
3. Software Development Process	30
4. Human Computer Interaction Principles	20
5. VB.NET programming environment	50
6. Develop and test a VB.NET application	50
TOTAL	200

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Fundamentals of Information Systems	1.1 Information systems 1.1.1 Definition 1.1.2 Components 1.2 Types of information systems 1.2.1 Transaction Processing Systems 1.2.2 Management Information Systems 1.2.3 Decision Support Systems 1.2.4 Executive Information Systems 1.2.5 Office Automation Systems 1.3 Emerging trends in information systems 1.4 Recommendation of information systems for different scenarios 1.5 Information system security 1.5.1 Definition 1.5.2 Information security management system 1.5.3 Tools for information system security 1.5.4 Firewalls 1.5.5 Virtual private networks 1.6 Mobile security 1.6.1 Geolocation software 1.6.2 Remote data removal software	• Oral questioning • Written tests • Practical tests

	1.7 Web security 1.7.1 Cyber security 1.7.2 Technologies 1.7.3 Web threats 1.7.4 Defence strategies	
2. Security measures to data, hardware, software in automated environment	2.1 Data security and control 2.2 Security threats and control measures 2.3 Types of computer crimes 2.4 Detection and protection against computer crimes 2.5 Laws governing protection of ICT	• Written tests • Oral questioning • Project • Practical tests
3. Software Development Process	3.1 Software Development Life Cycle 3.2 Software Development Methodologies 3.2.1 Waterfall 3.2.2 Spiral 3.2.3 Rapid Application Development 3.2.4 Agile Development 3.3 Modelling techniques 3.3.1 Data Flow Diagrams 3.3.2 Entity Relation Diagrams 3.3.3 UML diagrams 3.4 Creation of models for given scenarios	• Written tests • Oral questioning • Practical tests
4. Human Computer Interaction Principles	4.1 Human Computer Interaction 4.1.1 Definition 4.1.2 Role of interaction design 4.1.3 Interaction styles 4.1.4 Interaction elements 4.1.5 Mistakes in interaction design	• Practical • Oral questioning • Observation • Written tests

	4.2 Interface design principles 4.3 Prescribing interaction choices and recognition of interaction flaws	
5. VB.NET programming environment	5.1 The .Net framework 5.1.1 Applications supported 5.1.2 Components of the .Net framework 5.2 Installation of Visual Studio 5.3 Features of VB.Net 5.4 The Integrated Development Environment (IDE) 5.4.1 Definition of IDE 5.4.2 Parts of VB.Net IDE 5.5 VB.Net program structure 5.5.1 VB.NET syntax 5.5.2 Namespace declaration 5.5.3 Class or module 5.5.4 Procedures 5.5.5 Data types, variables, constants 5.5.6 The Main procedure 5.5.7 Statements and Expressions (Variable declarations, operations, control statements) 5.5.8 Comments 5.6 Creating aVB.Net project 5.6.1 Saving Forms and Project 5.6.2 Compiling a Project	• Practical tests • Oral tests • Written tests
6. VB.NET application development and testing	6.1 Basic VB.Net Controls 6.1.1 test Controls and their purpose	• Practical tests • Oral tests

	6.1.2 Standard naming conventions for controls 6.2 Elements of a control 6.2.1 Properties 6.2.2 Methods 6.2.3 Events 6.3 Demonstrating Properties, Methods and Events 6.3.1 Properties for basic controls 6.3.2 Setting properties at design time and run time 6.3.3 Methods for basic controls 6.3.4 Events for basic controls 6.4 Demonstrating event handling 6.4.1 Mouse events 6.4.2 Keyboard events 6.5 Designing VB.NET form using HCI principles 6.6 Connection of VB.Net applications to a database 6.6.1 ADO.Net object model 6.6.2 Demonstrating Database connection using the Data Provider 6.6.3 Demonstrating creation of tables using Dataset components 6.7 Deployment of VB.NET VB.Net applications 6.7.1 Purpose deployment 6.7.2 Demonstrating deployment steps	• Written tests
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Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised practical assignments and projects;
- Visiting expert from the ICT sector;
- Industrial visits

Recommended Resources for 25 trainees

S/No.	Category	Item Description / Specifications	Quantity	Recommended Ratio (Trainee: Item)
1.	Learning Materials	Textbooks	25	1:1
2.		Printed handouts	25 Sets	1:1
3.		User guides for Visual Studio and VB.NET programming	25	1:1
4.		Access to e-books, online platforms	25 Logins	1:1
5.	Learning Facilities	ICT Lab with desks, chairs, projector, whiteboard	1 Room	Shared
6.	Infrastructure	Internet access	1 Setup	Shared
7.		Power backup system	1 System	Shared
8.	Tools & Equipment	Computers with Visual Studio & SQL Server installed	25	1:1
9.		External storage devices	25	1:1
10.		Visual Studio Community Edition	Installed	N/A
11.		Antivirus/firewall	Installed	N/A
12.	Consumable Materials	Notebooks, pens, printed worksheets	25 Sets	1:1
13.		Project submission sheets, software license logs	25 Sets	1:1

ARTIFICIAL INTELLIGENCE CONCEPTS

ISCED UNIT CODE: 0613 554 12A

UNIT CODE: ICT/CU/CS/CC/04/6/MA

Relationship to Occupational Standards

This unit addresses the unit of competency: Implement Artificial Intelligence Concepts

Duration of Unit: 180 hours

Unit Description

This unit covers the competencies required to understand artificial intelligence. It involves understanding fundamentals of Artificial Intelligence, understanding problem solving techniques, understanding Python programming environment and developing Artificial Intelligence programs using Python.

Summary of Learning Outcomes

Learning outcomes	Duration (hours)
1. Fundamentals of Artificial Intelligence	30
2. Problem solving techniques	40
3. Python programming environment	60
4. Artificial Intelligence programs using python	50
TOTAL	180

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Concepts of Artificial Intelligence	1.1 Fundamentals Artificial Intelligence 1.2 History of Artificial Intelligence 1.3 Foundations of Artificial Intelligence	• Oral tests • Written tests • Practical tests

	1.3.1 Mathematics 1.3.2 Economics 1.3.3 Decision Theory 1.3.4 Neurology 1.3.5 Engineering 1.3.6 Psychology 1.3.7 Computer Networking 1.4 Applications of Artificial Intelligence 1.4.1 Expert systems 1.4.2 Machine Learning 1.4.3 Natural Language Processing 1.4.4 Gaming 1.4.5 Artificial Neural Networks 1.4.6 Computer Vision 1.5 Intelligence agents 1.6 Recognising Artificial Intelligence applications in real life	
2. Problem solving techniques	2.1 Logical operators 2.1.1 AND 2.1.2 OR 2.1.3 NOT 2.2 Propositional Logic and Predicate logic 2.3 Types of inferencing 2.3.1 Single Inferencing 2.3.2 Multiple inferencing 2.3.3 Case based reasoning 2.4 Machine Learning 2.5 Types of Machine Learning	• Oral tests • Written tests • Practical tests •

	2.5.1 Supervised Machine Learning 2.5.2 Unsupervised Machine Learning 2.6 Recognising applications of different types of inferencing	
3. Python programming environment	3.1 Installation of Python 3.1.1 Downloading Python Set Up 3.1.2 Running Python Set Up 3.2 Python syntax 3.2.1 The Zen of Python 3.2.2 Python Enhancement Proposals 8 (PEP 8) 3.2.3 Variable declaration. 3.2.4 Commenting 3.3 Python data types 3.3.1 Integer 3.3.2 Float 3.3.3 Boolean 3.3.4 Set 3.3.5 Dictionary 3.3.6 Tuple 3.3.7 List 3.3.8 String 3.4 Control structures in Python 3.4.1 Selection 3.4.2 Looping 3.5 Functions in Python 3.5.1 Built-in functions 3.5.2 User defined functions 3.5.3 Lambda functions 3.6 Object Oriented Python	• Oral tests • Written tests • Practical tests

	3.6.1 Creation of classes 3.6.2 Class variables 3.6.3 Class methods 3.7 Scientific Modules in Python 3.7.1 Pandas 3.7.2 Numpy 3.7.3 Matplotlib 3.8 Creation of applications using Scientific Modules	
4. Artificial Intelligence programs development using python	4.1 Sci-Kit Learn 4.2 Machine Learning with K-Nearest Neighbours 4.2.1 Mathematics behind K-Nearest Neighbours 4.2.2 Making Predictions with K-Nearest Neighbours 4.3 Machine Learning with Naïve Bayes Algorithm 4.3.1 Mathematics behind Naïve Bayes Algorithm 4.3.2 Making predictions with Naïve Bayes Algorithm	• Oral tests • Written tests • Practical tests

Suggested Methods of Instruction

- Presentations and practical demonstrations by trainer;
- Guided learner activities and research to develop underpinning knowledge;
- Supervised practical assignments and projects
- Visiting lecturer/trainer from the Computer Science sector;
- Industrial visits.

Recommended Resources for 25 trainees

S/No.	Category	Item Description / Specifications	Quantity	Recommended Ratio (Trainee:Item)
1.	Learning Materials	Textbooks	25 Books	1:1
2.		Handouts/slides	25 Sets	1:1
3.		Online course subscriptions for AI and Python	25 Licenses	1:1
4.		Python documentation and cheat sheets (print or digital)	25 Copies	1:1
5.	Learning Facilities	Classroom / ICT Lab with whiteboard, projector, speakers	1 Room	Shared
6.		Internet-enabled environment	1 Setup	Shared
7.	Infrastructure	High-speed internet connection	1 Connection	Shared
8.		Power backup	1 System	Shared

9.	Tools & Equipment	Laptops or desktop computers (Python 3.9+ compatible)	25 Units	1:1
10.		Pre-installed Python	Installed	N/A
11.		Libraries	Installed	N/A
12.		External storage	25 Devices	1:1
13.	Consumable Materials	Notebooks, pens,	25 Sets	1:1
14.		Assessment forms, quizzes, worksheets	25 Sets	1:1